

A cluster of probability problems in $\mathbb{R}^n$

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September 2024

Abstract

This is a note on n geometric probability problems that have a joint solution.

todo: add caveat about the meaning of ‘random’ in these maybe up here as a footnote, and remove these caveats from individual items

- Let V be a uniformly random k -dimensional subspace of $\mathbb{R}^n$.¹ What’s the probability that V contains a vector all of whose coordinates are positive?
- Take the standard hyperoctahedron in dimension n .² Slice through it with a random k -dimensional subspace; the shadow it leaves on the subspace is a k -dimensional polytope. How many faces of each dimension does this polytope have?
- What’s the probability that the convex hull of ℓ random vectors in $\mathbb{R}^m$ contains the origin?³
- What’s the expected fraction of the unit sphere $S^{m-1} \subseteq \mathbb{R}^m$ to lie in the positive cone spanned by ℓ random vectors?
- Take ℓ_b random blue vectors and ℓ_r random red vectors in $\mathbb{R}^m$. What’s the probability that there is a hyperplane through the origin with all red vectors on one side and all blue vectors on the other?

prob that convex hull contains origin (3b1b putnam problem), prob that positive cone contains origin, prob that all pts in some common half-space, prob that n random blue and m random red points can be separated with a hyperplane through the origin (the trick here is to note that this is equivalent to the blue and reflections of red pts being on the same side of a hyperplane)

¹Concretely, pick k vectors independently uniformly at random from $S^{n-1} \subseteq \mathbb{R}^n$ and let V be their span. (Or, equivalently, let the coordinates of these vectors be i.i.d. gaussians.) In case this isn’t formal enough for you — I mean that the probability measure is the uniform measure on the Grassmannian $\text{Gr}_k(\mathbb{R}^n)$: https://en.wikipedia.org/wiki/Grassmannian#Associated_measure

²The standard hyperoctahedron in dimension n is the convex hull of the standard basis and its reflection across the origin, i.e., with $e_1, \dots, e_n$ being the standard basis, the convex hull of the $2n$ vectors $\pm e_1, \pm e_2, \dots, \pm e_n$.

³This is a generalization of 1992 Putnam A6, <https://kskedlaya.org/putnam-archive/1992.pdf>, which asks for the probability that 4 vectors in $\mathbb{R}^3$ have the origin in their convex hull; 3Blue1Brown has a video on that problem: <https://youtu.be/0kmNXy7er84>.